

Tools and literature for completing the prospectus engine

7 October 2026. Code baseline: a05e18bbdeefbf278d29a6755e274124f97bda35, feature/prospectus-evidence-master.

We should adopt better source accounting, explicit document versions and stronger invariant tests within the existing engine. We should then run three bounded trials: Docling for document structure, INCEpTION for annotation and adjudication, and QuantLib for schedules and day counts. Replacing the legal engine or installing several new OCR systems would leave the most consequential defects unresolved.

The reason is visible in the BASF repair. The engine could execute the selected coupon calculation while an intermediate construction wrongly removed its common reference-period definition. A marginal instruction on printed p112 established that the definition applied to every Actual/Actual option. More solver power would not have recovered a premise that construction had deleted. The supplement's two printed incorporation ranges, 195-209 and 209-290, present a different problem: both can be transcribed perfectly while their legal relationship remains unresolved.

The [current repair result](#) records 77 BASF operations and 83 unresolved brackets. It also distinguishes installed-product integration from full construction, interpretation, settlement and independent acceptance. This survey preserves those distinctions. No new parser, annotation platform, solver backend or financial library was evaluated on our prospectuses during the survey.

Decision	Component or method	Gap it addresses	Local acceptance evidence required
Adopt the design	Web Annotation selectors, Akoma Ntoso document identities, PROV relationships	Ambiguous occurrences, editions, amendments and transformation history	Exact occurrence/edition round trips; explicit unresolved alternatives; justified exclusion and override accounting
Reuse installed tools	Z3 and Hypothesis	Correlated choices, finite predicates, controller and financial invariants	Agreement with the enumerating evaluator on a declared Boolean corpus; source-bound counterexamples; mutation and state-sequence checks
Adopt the evaluation method	ContractNLI evidence groups, SARA's separation of extraction and execution, DocLayNet split discipline	Missing distant exceptions, hidden defaults, leakage between related documents	Reviewed-evidence versus automatic-evidence arms; issuer/template/amendment-family separation; independent adjudication
Trial first	Pinned Docling sidecar alongside retained Poppler/PyMuPDF output	Columns, margins, lists, tables, reading order and text/box alignment	Every designated governing margin, footnote, number, negation and reference survives with correct attachment
Trial after occurrence mapping	INCEpTION with an explicit export bridge	Independent labels and review workflow	Lossless evidence-group round trip, blind assignments, recommendations disabled, separate adjudication

Decision	Component or method	Gap it addresses	Local acceptance evidence required
Trial after conventions are supplied	QuantLib v1.38 comparator	Coupon schedules, stubs, leap years and business-day adjustments	Independently derived dates/fractions and explicit source conventions; exact local monetary rounding retained
Borrow concepts; defer migration	ACTUS and FINOS CDM	Financial event/state and transfer representation	A demonstrated missing interoperability requirement, semantic mapping and targeted execution tests
Keep external evidence obligations	Primary legal/event sources and qualified independent readers	Actual applicability, exercise, client facts and acceptance	Dated, instrument-bound records and genuine independent readings

The [adoption program](#) turns these recommendations into ordered implementation and trial obligations. The [reading ledger](#), [source manifest](#) and [environment inventory](#) preserve the supporting material.

Preserve the selected contract before extending its reasoning

The current [occurrence binder](#) already requires a source hash, document, unit, start/end positions and an exact quote. That is a sound starting point. The missing part is complete accounting for why source occurrences survive, are excluded, are replaced, or remain unresolved in a particular construction. In [construction_ast.py](#), the renderer follows only the selected branch and records exclusions by option name. Coverage is accumulated mainly by visited Text nodes and Field templates. An excluded or overridden occurrence can therefore remain unaccounted for; a shared definition also needs a disposition that respects all of its uses.

The W3C [Web Annotation Data Model](#), §§4.2.4–4.2.5 and §4.3, supplies complementary text-quote and position selectors and a way to identify a representation’s state. Use these ideas in our existing JSON format: keep the immutable source edition, half-open character interval, quote and surrounding context, plus page geometry and the relevant construction operation. Do not silently replace the current exact binder with a generic annotation match. The standard permits normalized text and recommends treating multiple quote matches as multiple matches. A legal occurrence must remain ambiguous until its particular location is established. Normalized display text needs a separate map back to raw text.

There is a concrete interoperability trap. Web Annotation counts Unicode code points; INCEPTION’s TSV format counts UTF-16 code units. A supplementary-plane character occupies one position in Python strings and two units in that TSV format. The adapter must explicitly convert offsets, reject boundaries inside a surrogate pair, and check the recovered quote. Combining characters, line endings, hyphenation and duplicate clauses need separate tests. A common string is not an occurrence identifier.

[Akoma Ntoso 1.0](#), §4.1.3 and §5.7, separates the legal work, its expression or version, its physical/electronic manifestation and an individual copy. It also represents amendment events, validity and efficacy. Those distinctions fit our need to identify a prospectus, its language/version, the PDF bytes and subsequent supplements without conflating them. Adopt the identity and amendment concepts, with explicit governing-language and precedence premises; retain the current JSON rather than introducing an XML database. Akoma Ntoso assumes identifiable dated amendment events. Where their date or effect is disputed, preserve the dispute. It does not determine which BASF range is legally operative or supply our separate knowledge and observation times.

[PROV-DM](#), §§2.1 and 5.1–5.3, distinguishes entities, activities, agents, usage, generation and derivation. These names clarify the relation between source bytes, OCR, a corrected transcription, a construction decision and its reviewer. Its own discussion cautions that use and generation alone do not establish derivation. Our logs must record the actual transformation and source intervals. An attribution record is also not authentication of a reviewer’s identity. Full RDF/PROV serialization can wait for an exchange need.

The engineering condition is precise: within the admitted question and document scope, every relevant raw occurrence must have a compatible terminal disposition or remain explicitly unresolved. Every emitted character must be copied from a bound occurrence or attributed to a declared transformation, field value or separator. Excluding one branch must not exclude a definition shared by another. This condition detects missing accounting; it does not establish that the admitted scope or a supplied legal exclusion is correct.

Document tools: test the features that have actually failed

The existing OCR installation is sufficient to start. The application venv contains Hypothesis 6.151.9 and Z3 4.16.0.0; the pinned OCR interpreter contains PyMuPDF 1.27.2.2 and ResearchAssistant 0.1.0. They are different Python environments even though their executable symlinks resolve to the same base Python. The four retained OCR outputs still match their admitted hashes. Their English `eng.traineddata` is retained. Absence of a standalone tesseract executable does not negate the existing embedded OCR results. It also does not establish accuracy on Portuguese or German text: the current retained OCR profile accepts only the recorded 300-dpi English configuration.

The [previous Docling technical/code review](#) supports an isolated structure-preserving trial. Its report, [arXiv:2408.09869v5](#), §§3–4 and the output appendix, describes token geometry, layout regions, reading order and assembly. The inspected implementation at commit `570f956792fcfe8e51bde0f4a79a2ddabc1a061a` also has merge/dehyphenation paths whose original-text mappings need checking. The report’s 225-page timing evaluation used arXiv papers and IBM Redbooks with OCR off. It is not a prospectus accuracy result.

Keep the baseline’s raw words and boxes, and add paragraph, language, column, list/table and margin-attachment relations as a separate layer. Export that layer into [source_graph.py](#), then compare the same clause graph and questions with either baseline or candidate structure. Do not feed only exported Markdown into the legal engine. The first trial should include BASF p112 and supplement p12; Deutsche’s bilingual terms; the four retained BES extractions; and the BBVA/SEB financial-profile pages. Review the evidence groups before comparing candidate outputs.

Two new benchmark readings explain how to evaluate this trial. Pfizmann et al.’s [DocLayNet](#), §§3–5, uses 80,863 pages and eleven layout classes, with redundant annotation on part of the corpus. Financial reports and laws/regulations are included, but the corpus is predominantly English and avoids scans where possible. Its document-level split experiment found that a naive page split inflated the authors’ layout results by roughly ten mAP points. The [official dataset documentation](#) also distinguishes image-coordinate annotations from PDF-derived text-cell coordinates. Borrow the split and annotation discipline. Add governing-margin and clause-scope relations, which are not supplied by the layout labels.

[OmniDocBench](#), §§3–5 and supplementary §V, separates text, tables, formulas and reading order, using normalization and fuzzy matching to compare heterogeneous outputs. The retained manuscript describes 981 pages; later dataset versions must be treated separately. Crucially, its ignore handling excludes headers, footers, page numbers, page footnotes and captions on specified metrics. The inspected [official matching code](#), commit `f133a71e9e91c3621c7ce8994200a7b394a06eb3`, filters these categories too. That is a reasonable benchmark scope but an unsafe acceptance rule for BASF. Borrow the decomposition and matching diagnostics, while checking legally governing margins, footnotes, repeated punctuation and

references explicitly. Aggregate edit distance remains explanatory; it cannot excuse one lost exception.

MinerU and olmOCR remain alternatives to screen if the first trial fails on a named layout family. Their current official READMEs were inspected, not their technical papers or inference implementations. [MinerU’s README](#) describes CPU/accelerated paths, a document-library parse default of ten pages, and an Apache-based license with additional conditions. A future adapter must consume continuation locators and check total page coverage. [olmOCR’s README](#) describes a 7B VLM, local GPU requirements and past corrections to blank-page hallucinations and rotation. Neither has local accuracy evidence here. Marker, PaddleOCR, OCRmyPDF and GROBID were not technically audited in this extension; there is no basis to claim that they outperform the current path. ResearchAssistant can continue acquiring and parsing literature without making them production dependencies.

Rules, law and evidence require different checks

The current [predicate evaluator](#) enumerates completions over at most twelve Boolean facts. Z3 is already a pinned dependency. It can support larger correlated choice constraints and give concrete satisfying assignments or contradictory named constraints. That is a focused extension; it does not require replacing RuleIR, Catala or the legal service.

The proposed Boolean adapter has a small, checkable argument. Let C contain the admitted observations and choice constraints and let q be the translated query. Handle explicitly conflicting observations before simplifying q, as the existing evaluator does. If C is unsatisfiable, report a conflict or reject an invalid construction admission; never obtain a vacuous YES. Otherwise, q holds in all admissible worlds exactly when C AND NOT q is unsatisfiable, and fails in all worlds exactly when C AND q is unsatisfiable. Satisfiability of both yields UNKNOWN with two witnesses. This follows directly by taking the satisfying assignments as the admissible worlds. Translation of Boolean constants, fact names, negation, conjunction and disjunction must preserve their truth tables recursively.

This argument concerns the encoded finite Boolean problem. It supplies neither a legal interpretation nor a proof of our future adapter. Any required solver call returning unknown or timing out must preserve an unresolved result with the computational reason. The [official Z3 guide](#) and the retained installed API source document these outcomes. The existing enumerator is the comparator on its supported domain; unconstrained inputs must retain its YES/NO/UNKNOWN/CONFLICT behavior. A satisfying model is not an actual event, and an unsat core is not automatically minimal.

The prior [ContractNLI/SARA review](#) remains directly useful. ContractNLI (Koreeda and Manning, 2021, §§2–5 and Appendix A.1) requires self-contained evidence, including distant and discontinuous exceptions. Its reviewed-evidence comparison separates retrieval from inference. Its 607 NDAs and seventeen hypotheses, with scanned/multicolumn PDFs excluded, do not validate a bond classifier. SARA information extraction (Holzenberger and Van Durme, 2023, §§3–5, Algorithms 1–2 and Appendices A–F) supplies typed event/argument spans before an explicit statute program executes. Its empty-input control and date defaults warn against obtaining answers from missing facts.

Adopt evidence groups containing the governing clause, definition, condition, exception, amendment and applicable priority source. Compare execution on reviewed groups with execution on automatically assembled groups, using the same inference method and facts. If both arms fail, investigate translation or reasoning. If only automatic input fails, investigate extraction and reference closure. Do not convert ContractNLI’s “not mentioned” label into proof of contractual absence.

Catala, LegalRuleML and structured argumentation remain useful within those boundaries. The retained Catala paper’s compilation argument concerns an explicitly defined calculus, not the

translation from a prospectus. LegalRuleML represents alternative interpretations and their contexts. Prakken’s argumentation framework distinguishes attacks on premises, conclusions and inferences, with justified preferences affecting defeat. None authorizes priority based on proximity, model confidence or an unexplained “later text wins” rule. Preserve unresolved priority and cycles explicitly; use the existing formal backends only after translation.

[law_facts.py](#) already separates effective, knowledge and observation dates. Its results still say actual exercise is not established. Akoma Ntoso identities can improve the legal-source registry, but actual closure requires the applicable statute and commencement or transitional provision, regulator order, operative judgment and any appeal/stay, issuer notice, and entity/security identifiers. The 25 hypothetical law scenarios remain development tests. An open-ended validity interval cannot certify current law. A jurisdiction adapter should record the authority, edition and known publication history, then leave missing applicability or event facts UNKNOWN.

Financial tools are comparators, not missing contractual premises

QuantLib is the most useful first financial trial. The earlier schedule review found convenience defaults for evaluation dates, business-day conventions and termination conventions. This extension inspected v1.38 [ActualActual](#), [day-counter tests](#) and the [TARGET calendar](#). The implementation distinguishes schedule-based ISMA/ICMA from an older path that estimates reference periods when they are absent. The tests cover short/long coupons and also show that ISDA, ISMA and AFB can give different fractions for the same dates. The calendar implements specific holidays; it is not a live source of every local or exceptional closure.

Require the schedule generation rule, first/last stub, end-of-month rule, holiday calendar edition, adjustment conventions, accrual basis and reference periods before calling the comparator. Check actual payment dates and source-derived fractions, then compare exact rounded amounts with the local kernels. A missing reference period must prompt an unresolved convention, not trigger the older fallback. Full settlement additionally needs entitlement/record and ex-coupon rules, event ordering, notices, currency/FX basis, taxation where in scope, holder aggregation, and a complete cash/security transfer trace. None is supplied merely by a matching year fraction. Only the schedule/day-count/calendar slice has received source inspection here; QuantLib’s full settlement machinery has not.

The [ACTUS specification](#), retained v1.1, §§2.4–2.10, §3, §§4–6 and §7.1’s principal-at-maturity tables, makes useful distinctions between contract terms, changing state, scheduled/unscheduled events, payoff functions and state transitions. Equal-time event ordering matters. Borrow this separation for an explicit financial event trace. Its conventions also require care: the specification has no timezone support, supplies a NoHoliday default, and obtains external events/risk factors through observer interfaces. These are inputs, not discovered legal facts.

Defer adopting the ACTUS runtime. The retained [service build](#) requires Java 17 and `actus-core:1.1.0`, while its [README](#) contains older webapp instructions and says to request an access token for the core. The core itself was not audited or executed. Its [custom license](#), particularly §§2.3–2.5, also contains conformance, attribution and distribution terms distinct from the specification’s CC BY-SA license. Runtime access, the intended use and the exact version need resolution before integration; the public wrapper is insufficient.

FINOS CDM supplies an exchange model rather than a prospectus reader. The retained [settlement types](#) require asset, quantity, units, payer/receiver and settlement date, with explicit cash/physical choices. Its [bond execution example](#) includes delivery versus payment, security identity, counterparties and clean-price/accrued-interest components. These are useful mapping requirements. The example supplies the accrued component; it does not derive our selected bond’s legal entitlement. Defer a complete CDM adapter until a bank/system needs that exchange format. No Rosetta runtime, full event calculus or round-trip implementation was tested.

Test changing evidence and obtain independent readings

Hypothesis is already installed and pinned. Its [stateful testing documentation](#) and retained implementation support sequences of actions against a simpler model, with invariants checked between steps. Use it for construct-amend-withdraw-query sequences and execute-mutate-interrupt-recover-replay sequences. Predeclared mutations should include a lost margin, moved negation, repeated occurrence, wrong source edition, changed required legal premise, identical holder names with different identities, changed language data, corrupt product and relevant or irrelevant input change. Store the minimized failures as permanent examples. Generated tests cannot expose a legal obligation omitted from both models.

The retained CUTEcat paper (Goutagny et al., 2025, §§2-5) adds branch-directed concolic testing and exception-order reduction. Its larger example generated 186,390 cases in 6h37m. Borrow the mutation and branch-coverage ideas now; defer its separate execution image until a concrete Catala path needs it. The existing reading inspected the official artifact README/license, not the complete runtime. The [earlier broader survey](#) also covers Stipula, eFLINT, Symboleo and ContractCheck. They offer useful lifecycle or consistency models but have different semantic domains and bounds. Nothing inspected justifies migrating the prospectus service wholesale.

Build Systems à la Carte (Mokhov, Mitchell and Peyton Jones, 2018, §§3-6) supplies the controller design: separate scheduling from the decision to rebuild and bind actual dependencies, task definitions and results. The [previous technical and official-code review](#) includes the paper's hidden-read and nondeterminism qualifications. [controller.bindings](#) currently hashes broad source and environment sets. That is conservative for declared dependencies, but it neither proves absence of hidden reads nor avoids unnecessary invalidation. Add an explicit read capability and immutable phase inputs; test filesystem, environment, subprocess and clock inputs separately. A Python wrapper alone cannot establish that native tools have no hidden reads. Use process isolation or retain that limitation before claiming hermetic execution.

INCEpTION (Klie et al., 2018, §§2-4) supports typed annotation, relations, curation and annotation assistance. The retained [v38 user guide](#), curation section, Appendix A and Appendix D, adds material integration constraints. Anonymized curation hides names from curators but not managers and does not randomize the order. Discontinuous spans have no immediate native support; the guide suggests linking separate annotations. TSV is UTF-8 text with UTF-16 offsets. The inspected [TSV adapter](#) even marks the format as prone to inconsistencies. Its code snapshot is a later 43.0-SNAPSHOT, not the v38 guide's release. Pin a matched release, documentation and export format before a trial. The v38 guide says Java 11 or higher; this is not evidence that every later release runs on our Java 11 installation.

Use separate read-only source packets and reviewer assignments. Disable recommendations, shared model explanations and automatic vote-based pre-merging for independent acceptance. Preserve each reader's labels and evidence groups before adjudication by a separately assigned reviewer. Measure disagreement on meaning and source scope as well as labels. INCEpTION may make this work easier; it cannot supply qualified readers, verify their independence or turn a majority vote into legal authority. An exportable local form remains the fallback if its bridge cannot preserve the evidence.

Recommended order and limits of this survey

First finish occurrence/disposition accounting and source-backed reference closure, including the BASF common definition. Introduce the Z3 comparator and Hypothesis invariants around that representation. Then test document structure on frozen development pages, complete the financial convention/event trace, and exercise actual dependency changes. Prepare the annotation bridge early enough to freeze independent labels before acceptance results are shown. P7/P8 remain blocked until real bound labels, scope and sign-offs exist.

Decision	Primary evidence status	Veto or uncertainty	Next justified action	What is not established
Complete source accounting	Mechanism and current code mismatch inspected	Legal selection and priority still supplied	Implement interval dispositions and shared-definition regressions	Complete contract construction
Reuse Z3 and Hypothesis	Installed versions and relevant APIs inspected	New adapter and tests not yet implemented	Compare against finite enumeration and an independent state model	Correct natural-language translation
Trial Docling, INCEpTION and QuantLib	Relevant methods, implementation slices and limitations inspected	No local candidate evaluation	Run each predeclared trial after its input prerequisites	Tool superiority or readiness as a default
Defer ACTUS and CDM runtime changes	Relevant specification and wrapper/model slices inspected	Core access and version; additional integration scope	Revisit only for a demonstrated event or exchange requirement	Complete settlement or regulatory approval
Continue independent acceptance work	Existing admission interfaces available	Real independent labels and actual facts absent	Supply and adjudicate genuine evidence	Release acceptance

ResearchAssistant’s repaired metadata search used eighteen provider requests, yielding 96 provider rows and 45 distinct records, with twelve automatically nominated leads. The broad wording retrieved mostly blockchain, unrelated contracts and general prospectus law. All nominations have explicit dispositions in [discovery-disposition.json](#); none is used as technical adoption evidence. The earlier discovery failed its aggregate-record limit and lost its partial bundle; it is conservatively charged as 33 possible requests. Exact-title/official-source acquisition and the retained technical literature provide the substantive coverage. The manifest records failed URLs, version distinctions and the bounded retrieval count. This is a thorough review of the named engineering gaps, not a systematic-review claim of literature saturation or proof that no other useful tool exists.

The strongest alternative explanation is that an untried document model could solve more of the extraction problems than expected. The paired Docling trial can test that without weakening source requirements. The weakest present evidence is local comparative performance and independent legal interpretation; neither was measured here. The survey supports an adoption order and concrete tests, while leaving production promotion to their results.